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Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator

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The Unified Form Language (UFL) (Alnaes et al., 2013) is essential for writing variational forms of partial differential equations (PDEs) in automated finite element solvers like FEniCS. However, UFL has a critical limitation: non-trivial solid mechanics problems, such as complex plasticity or neural network constitutive models, cannot be naturally formulated within it. This restriction has slowed FEniCS adoption in the broader solid mechanics community.

To overcome this bottleneck, we introduce DOLFINx-External-Operator (Latyshev et al., 2025), a novel software framework for FEniCSx. Based on DOLFINx's new data-centric design (Baratta et al., 2023), automatic UFL Expression code generation, and the UFL ExternalOperator extension (Bouziani & Ham, 2021), this tool enables the seamless integration of arbitrary constitutive models via third-party packages that support ndarray-like data structures, such as JAX (Bradbury et al., 2018) and Numba (Lam et al., 2015).

As a particular outcome, we leverage JAX for forward-mode algorithmic automatic differentiation (AD), eliminating the need for error-prone manual derivatives. We demonstrate the use of AD on a complex non-associative Mohr-Coulomb elastoplastic model with apex smoothing (Helfer et al., 2024). The framework and examples are open-source under the LGPLv3 license.

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